Raffles Institution
H2 Mathematics (9758)
Solution for 2025 A-Level Paper 1

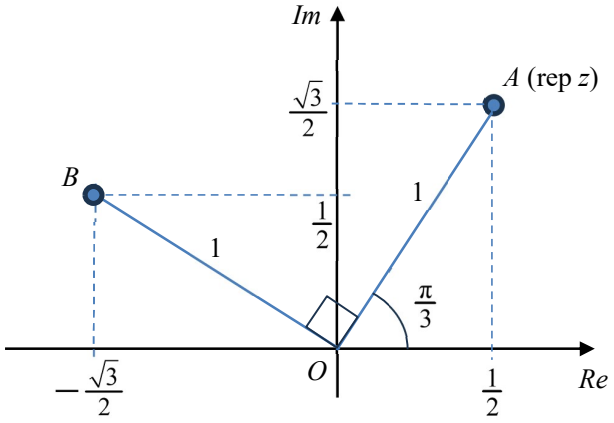
Question 1

No.	Suggested Solution	Remarks for Student
(a)	$u_1 = a$ $u_2 = 3u_1 + 5 = 3a + 5$ $u_3 = 3u_2 + 5 = 3(3a+5) + 5 = 9a + 20$	
(b)	$u_4 = 3u_3 + 5 = 3(9a+20) + 5 = 27a + 65$ Since $\sum_{r=1}^{4} u_r = 110$, $a + (3a+5) + (9a+20) + (27a+65) = 110$ $40a + 90 = 110$ $a = 0.5$	

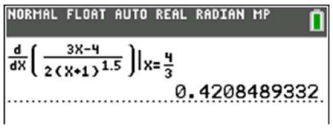
Question 2

No.	Suggested Solution	Remarks for Student
(a)	Let the first term be a and the common difference be d. $S_{20} = 65$ $\frac{20}{2}(2a + 19d) = 65$ $4a + 38d = 13$ --- (1) Sum of 21st to 28th term is -30: $S_{28} - S_{20} = -30$ $\frac{28}{2}(2a + 27d) - 65 = -30$ $28a + 378d = 35$ --- (2) Solving (1) and (2), we have $a = 8,\ d = -0.5$. $\therefore$ The first term is 8 and the common difference is -0.5.	
(b)	Since $\frac{u_n}{u_{n-1}} = \frac{e^{k(n+1)}}{e^{kn}} = e^{kn+k-kn} = e^k$ **is a constant** independent of n. Since the ratio of every two consecutive terms is a constant, the sequence is geometric. (shown)	As this is a "show" question, show clear working!

Question 3

No.	Suggested Solution	Remarks for Student
(a)	$z=\dfrac{\sqrt{3}+\mathrm{i}}{\sqrt{3}-\mathrm{i}}=\dfrac{\left(\sqrt{3}+\mathrm{i}\right)\left(\sqrt{3}+\mathrm{i}\right)}{\left(\sqrt{3}-\mathrm{i}\right)\left(\sqrt{3}+\mathrm{i}\right)}=\dfrac{\sqrt{3}^2+2\sqrt{3}\mathrm{i}+\mathrm{i}^2}{\sqrt{3}^2-\mathrm{i}^2}=\dfrac{2+2\sqrt{3}\mathrm{i}}{4}=\dfrac{1}{2}+\dfrac{\sqrt{3}}{2}\mathrm{i}$ Thus z is in the 1st quadrant, and we have $\lvert z\rvert=\sqrt{\left(\dfrac{1}{2}\right)^2+\left(\dfrac{\sqrt{3}}{2}\right)^2}=\sqrt{\dfrac{1}{4}+\dfrac{3}{4}}=1$ and $\arg(z)=\tan^{-1}\dfrac{\left(\frac{\sqrt{3}}{2}\right)}{\left(\frac{1}{2}\right)}=\tan^{-1}\sqrt{3}=\dfrac{\pi}{3}$	Note that the use of calculator is **not** allowed as stated in the question. Hence, clear working of the steps must be shown.
(b)	 Since a rotation through $\dfrac{\pi}{2}$ anticlockwise about the origin corresponds to a multiplication of a complex number by i, B represents the complex number $\mathrm{i}z$.	Positions of points A and B must be identifiable, ie indicate the Re and Im part of the complex numbers OR the modulus and argument of the complex numbers clearly.

Question 4

No.	Suggested Solution	Remarks for Student
(a)	$y=\dfrac{3x+10}{\sqrt{x+1}}$ $\dfrac{\mathrm{d}y}{\mathrm{d}x}=\dfrac{\sqrt{x+1}(3)-(3x+10)\frac{1}{2}(x+1)^{-\frac{1}{2}}}{x+1}$ $=\dfrac{(x+1)^{-\frac{1}{2}}\left[3(x+1)-\frac{1}{2}(3x+10)\right]}{x+1}$ $=\dfrac{\frac{3}{2}x-2}{(x+1)^{\frac{3}{2}}}$ $=\dfrac{3x-4}{2(x+1)^{\frac{3}{2}}}$ When $\dfrac{\mathrm{d}y}{\mathrm{d}x}=0,\quad 3x-4=0\Rightarrow x=\dfrac{4}{3}$ Thus the x-coordinate of the turning point is $\dfrac{4}{3}$.	Tip for **(b)**: Simplify the expression for $\dfrac{\mathrm{d}y}{\mathrm{d}x}$ to $\dfrac{3x-4}{2(x+1)^{\frac{3}{2}}}$ before differentiating again!
(b)	$\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2}=\dfrac{2(x+1)^{\frac{3}{2}}(3)-(3x-4)2\left(\frac{3}{2}\right)(x+1)^{\frac{1}{2}}}{4(x+1)^3}$ $=\dfrac{(x+1)^{\frac{1}{2}}\left[6(x+1)-3(3x-4)\right]}{4(x+1)^3}$ $=\dfrac{-3x+18}{4(x+1)^{\frac{5}{2}}}$ When $x=\dfrac{4}{3}$, $\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2}=\dfrac{-4+18}{4\left(\frac{4}{3}+1\right)^{\frac{5}{2}}}\approx 0.421>0$ Therefore, the turning point is a minimum point.	Note that the question says "use calculus to find the value of the second derivative", so this method is mandatory. You may use GC to check your answer nevertheless. 

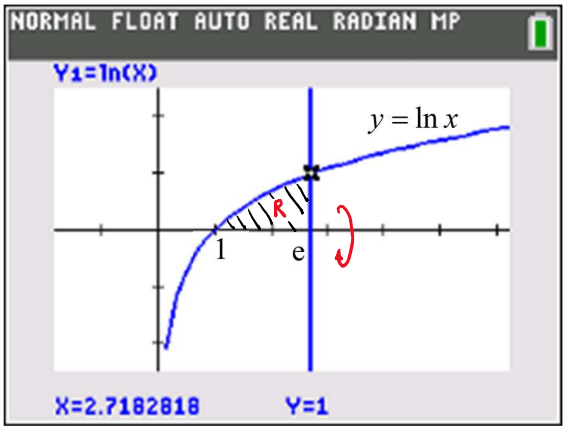
Question 5

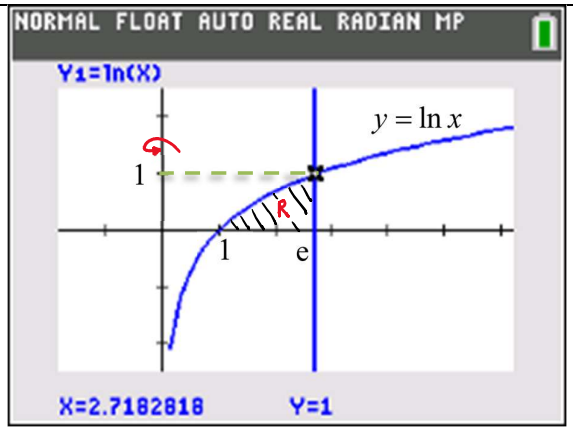
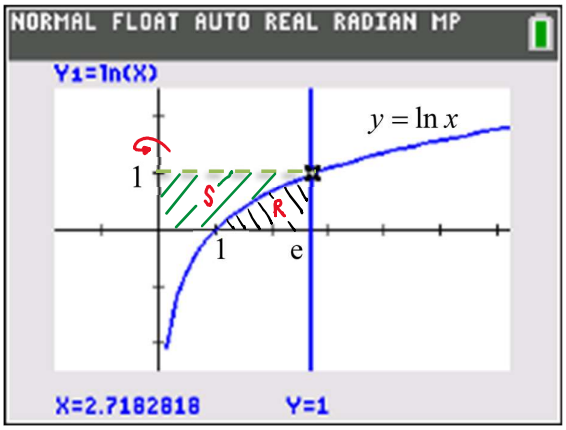
No.	Suggested Solution	Remarks for Student
(a)	$e^x \geq 2e^{-x}+1$ Let $e^x = m$, then we have $m \geq \frac{2}{m}+1$ Since m is always positive, $m^2 \geq 2+m$ $m^2-m-2 \geq 0$ $(m-2)(m+1) \geq 0$ $m \leq -1$ or $m \geq 2$ $e^x \leq -1$ $\quad$ $e^x \geq 2$ (n.a. since $e^x > 0 \ \forall x \in \mathbb{R}$) $\quad$ $x \geq \ln 2$	
(b)	From **(a)**, $e^x - 2e^{-x} - 1 \geq 0$ when $x \geq \ln 2$ We note also that $0 < \ln 2 < 1$. $\int_0^1 \left\lvert e^x - 2e^{-x} - 1 \right\rvert dx$ $= \int_0^{\ln 2} -\left(e^x - 2e^{-x} - 1\right) dx + \int_{\ln 2}^1 \left(e^x - 2e^{-x} - 1\right) dx$ $= \left[-\left(e^x + 2e^{-x} - x\right)\right]_0^{\ln 2} + \left[e^x + 2e^{-x} - x\right]_{\ln 2}^1$ $= \left[-\left(e^{\ln 2} + 2e^{-\ln 2} - \ln 2\right) + (1+2-0)\right] + \left[e^1 + 2e^{-1} - 1 - \left(e^{\ln 2} + 2e^{-\ln 2} - \ln 2\right)\right]$ $= \left[-\left(2 + 2\left(\frac{1}{2}\right) - \ln 2\right) + (1+2-0)\right] + \left[e^1 + 2e^{-1} - 1 - \left(2 + 2\left(\frac{1}{2}\right) - \ln 2\right)\right]$ $= e + \frac{2}{e} - 4 + 2\ln 2$	Recall that $\lvert f(x) \rvert$ $= \begin{cases} f(x), & f(x) \geq 0 \\ -f(x), & f(x) < 0 \end{cases}$ Tip: Use GC to check your answer!

Question 6

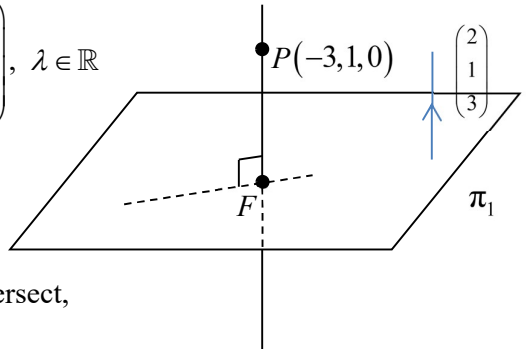
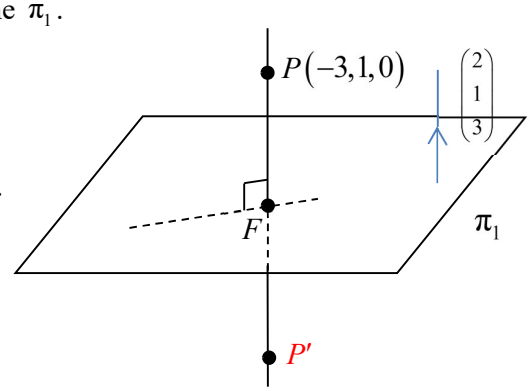
No.	Suggested Solution	Remarks for Student
(a)	$\dfrac{\mathrm{d}M}{\mathrm{d}t} = -kM$ $\dfrac{1}{M}\dfrac{\mathrm{d}M}{\mathrm{d}t} = -k$ Integrate w.r.t t, $\int \dfrac{1}{M}\mathrm{d}M = \int -k\ \mathrm{d}t$ Since $M > 0$, $\ln M = -kt + c \quad$ where $c \in \mathbb{R}$ $M = \mathrm{e}^{-kt+c}$ $= A\mathrm{e}^{-kt}$, where $A = \mathrm{e}^{c}$ When $t = 0$, $M = M_0$: $M_0 = A(1) \Rightarrow A = M_0$ Hence $M = M_0\mathrm{e}^{-kt}$ When $t = 5730$, $M = \dfrac{1}{2}M_0$: $\dfrac{1}{2}M_0 = M_0\mathrm{e}^{-k(5730)}$ $\mathrm{e}^{-k(5730)} = \dfrac{1}{2}$ $\mathrm{e}^{k(5730)} = 2$ $k = \dfrac{\ln 2}{5730}$	
(b)	When $M = 0.2M_0$, we have $0.2M_0 = M_0\mathrm{e}^{-\left(\frac{\ln 2}{5730}\right)t}$ $\mathrm{e}^{-\left(\frac{\ln 2}{5730}\right)t} = 0.2$ $-\left(\dfrac{\ln 2}{5730}\right)t = \ln 0.2$ $t = -\left(\dfrac{5730}{\ln 2}\right)\ln 0.2 = 13\ 304 = 13\ 300$ (nearest hundred) Therefore, the plant died 13300 years ago.	

Question 7

No.	Suggested Solution	Remarks for Student
(a)	$\frac{\mathrm{d}}{\mathrm{d}x}(x\ln x - x)$ $= x\left(\frac{1}{x}\right) + \ln x - 1$ $= \ln x$ (shown)	
(b)	 Volume generated when R is rotated 360° about the x-axis $= \pi\int_1^{\mathrm{e}} (\ln x)^2\,\mathrm{d}x$ $= \pi\left\{\left[x(\ln x)^2\right]_1^{\mathrm{e}} - \int_1^{\mathrm{e}} 2\ln x\,\mathrm{d}x\right\}$ $u = (\ln x)^2$, $\frac{\mathrm{d}u}{\mathrm{d}x} = 2\ln x\left(\frac{1}{x}\right)$; $\frac{\mathrm{d}v}{\mathrm{d}x} = 1$, $v = x$ $= \pi\left\{\left[\mathrm{e}(1)^2 - 0\right] - 2\left[x\ln x - x\right]_1^{\mathrm{e}}\right\}$ using result from **(a)** $= \pi\left\{\mathrm{e} - 2\left[\mathrm{e}(1) - \mathrm{e} - (0-1)\right]\right\}$ $= \pi(\mathrm{e}-2)$ units^3 **OR** $= \pi\int_1^{\mathrm{e}} (\ln x)^2\,\mathrm{d}x$ $= \pi\left\{\left[(\ln x)(x\ln x - x)\right]_1^{\mathrm{e}} - \int_1^{\mathrm{e}} (\ln x - 1)\,\mathrm{d}x\right\}$ $u = \ln x$, $\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{x}$; $\frac{\mathrm{d}v}{\mathrm{d}x} = \ln x$, $v = x\ln x - x$ $= \pi\left\{\left[0-0\right] - \left[(x\ln x - x) - x\right]_1^{\mathrm{e}}\right\}$ using result from **(a)** $= -\pi\left[(\mathrm{e} - 2\mathrm{e}) - (0-2)\right]$ $= -\pi\left[(-\mathrm{e}) + 2\right]$ $= \pi(\mathrm{e}-2)$ units^3	

(c)	 Volume generated when R is rotated 360° about the y-axis $= \pi \int_0^1 (e)^2 - (e^y)^2 \, dy$ $= 13.18 \text{ units}^3$ (2 d.p) **OR**  Volume generated when R is rotated 360° about the y-axis $=$ Volume of cylinder $-$ Volume generated when S is rotated about the y-axis $= \pi (e)^2 (1) - \pi \int_0^1 (e^y)^2 \, dy$ $= 13.18 \text{ units}^3$ (2 d.p)	Sketch the graph to help you visualize!

Question 8

No.	Suggested Solution	Remarks for Student
(a)	Let F be the foot of the perpendicular from P to the plane π_1. $l_{PF}: \mathbf{r} = \begin{pmatrix}-3\\1\\0\end{pmatrix} + \lambda\begin{pmatrix}2\\1\\3\end{pmatrix},\ \lambda \in \mathbb{R}$ $\pi_1: \mathbf{r}\cdot\begin{pmatrix}2\\1\\3\end{pmatrix} = 2$  When l_{PF} and π_1 intersect, $\left[\begin{pmatrix}-3\\1\\0\end{pmatrix} + \lambda\begin{pmatrix}2\\1\\3\end{pmatrix}\right]\cdot\begin{pmatrix}2\\1\\3\end{pmatrix} = 2$ $(-6+1)+\lambda(4+1+9) = 2$ $\lambda = \frac{1}{2}$ $\therefore \overrightarrow{OF} = \begin{pmatrix}-3\\1\\0\end{pmatrix} + \frac{1}{2}\begin{pmatrix}2\\1\\3\end{pmatrix} = \begin{pmatrix}-2\\1.5\\1.5\end{pmatrix}$ and the coordinates of F are $(-2,\ 1.5, 1.5)$.	Draw diagrams to help you visualize!
(b)	Let P' be the reflection of P in the plane π_1. [By addition of vectors] $\overrightarrow{OP'} = 2\overrightarrow{OF} - \overrightarrow{OP} = 2\begin{pmatrix}-2\\\frac{3}{2}\\\frac{3}{2}\end{pmatrix} - \begin{pmatrix}-3\\1\\0\end{pmatrix} = \begin{pmatrix}-1\\2\\3\end{pmatrix}$.  [OR By Ratio Theorem] $\overrightarrow{OF} = \frac{\overrightarrow{OP} + \overrightarrow{OP'}}{2} \Rightarrow \overrightarrow{OP'} = 2\overrightarrow{OF} - \overrightarrow{OP}$	
(c)	Given $\pi_2: \mathbf{r} = \begin{pmatrix}1\\4\\-1\end{pmatrix} + \lambda\begin{pmatrix}2\\-1\\5\end{pmatrix} + \mu\begin{pmatrix}2\\2\\3\end{pmatrix}$ A normal to π_2, $\mathbf{n}_2 = \begin{pmatrix}2\\-1\\5\end{pmatrix} \times \begin{pmatrix}2\\2\\3\end{pmatrix} = \begin{pmatrix}-13\\4\\6\end{pmatrix}$	Since $\begin{pmatrix}-13\\4\\6\end{pmatrix}$ is perpendicular to both $\begin{pmatrix}2\\-1\\5\end{pmatrix}$ and $\begin{pmatrix}2\\2\\3\end{pmatrix}$. Check that you have done the

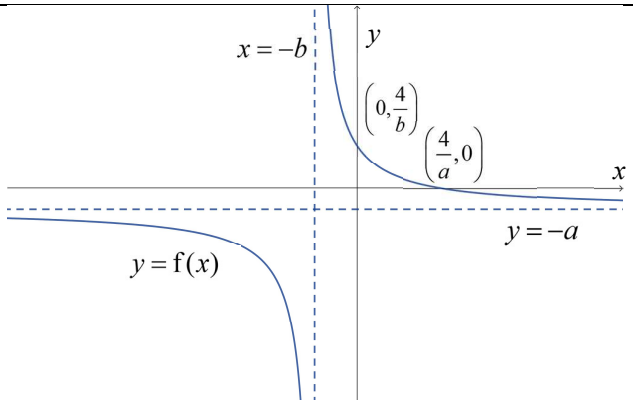
	Let the acute angle between the planes be θ. $\cos\theta = \dfrac{\left\|\begin{pmatrix}2\\1\\3\end{pmatrix}.\begin{pmatrix}-13\\4\\6\end{pmatrix}\right\|}{\sqrt{4+1+9}\sqrt{169+16+36}} = \dfrac{4}{\sqrt{14}\sqrt{221}}$ $\theta = 85.9^\circ$ (1 d.p)	cross product correctly by checking that $\begin{pmatrix}2\\-1\\5\end{pmatrix}\bullet\begin{pmatrix}-13\\4\\6\end{pmatrix} = \ldots = 0$ $\begin{pmatrix}2\\2\\3\end{pmatrix}\bullet\begin{pmatrix}-13\\4\\6\end{pmatrix} = \ldots = 0$

Question 9

No.	Suggested Solution	Remarks for Student
(a)	$y = \sin^{-1} px$ $\sin y = px$ Differentiate w.r.t. x, $\cos y \dfrac{dy}{dx} = p$ $\dfrac{dy}{dx} = \dfrac{p}{\cos y} = p\sec y$ (shown)	Rewrite the given question to $\sin y = px$ before differentiating!
(b)	From part **(a)**, $\dfrac{dy}{dx} = p\sec y$ Differentiate w.r.t. x, $\dfrac{d^2y}{dx^2} = p\sec y\tan y\dfrac{dy}{dx}$ $\dfrac{d^2y}{dx^2} = \tan y\left(\dfrac{dy}{dx}\right)^2$ since $\dfrac{dy}{dx} = p\sec y$ Differentiate w.r.t. x, $\dfrac{d^3y}{dx^3} = \tan y\cdot 2\left(\dfrac{dy}{dx}\right)\left(\dfrac{d^2y}{dx^2}\right)+\left(\dfrac{dy}{dx}\right)^2\sec^2 y\left(\dfrac{dy}{dx}\right)$ $= 2\tan y\left(\dfrac{dy}{dx}\right)\left(\dfrac{d^2y}{dx^2}\right)+\sec^2 y\left(\dfrac{dy}{dx}\right)^3$ When $x = 0,\ y = \sin^{-1} 0 = 0$ $\dfrac{dy}{dx} = p\sec 0 = p$ $\dfrac{d^2y}{dx^2} = (\tan 0)(p)^2 = 0$ $\dfrac{d^3y}{dx^3} = 2(\tan 0)(p)(0)+(\sec^2 0)(p)^3 = p^3$ The Maclaurin series for y is $y = 0+px+0x^2+\dfrac{p^3}{3!}x^3+\cdots$ $= px+\dfrac{p^3x^3}{6}+\cdots$	For this topic, keep a look out on how you may simplify the equations before differentiating again. Here, simplify $\dfrac{d^2y}{dx^2}$ to $\tan y\left(\dfrac{dy}{dx}\right)^2$ so that obtaining $\dfrac{d^3y}{dx^3}$ is not so complicated.

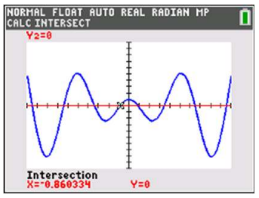
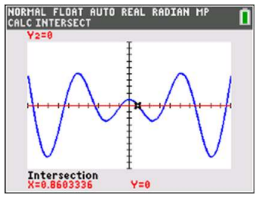
(c)	Observe that $y=\sin^{-1} px \Rightarrow \dfrac{\mathrm{d}y}{\mathrm{d}x}=\dfrac{p}{\sqrt{1-p^2x^2}}$ Let $p=3$. From part **(b)**, $\sin^{-1} 3x=3x+\dfrac{9}{2}x^3+\ldots$ Differentiate w.r.t. x, $\dfrac{3}{\sqrt{1-9x^2}}=3+\dfrac{27}{2}x^2+\ldots$ $\dfrac{1}{\sqrt{1-9x^2}}=1+\dfrac{9}{2}x^2+\ldots$	

Question 10

No.	Suggested Solution	Remarks for Student
(a)		
(b)	$y=\mathrm{f}(x)=\dfrac{4-ax}{x+b}=\dfrac{-a(x+b)+4+ab}{x+b}=-a+\dfrac{4+ab}{x+b}$ $y=-a+\dfrac{4+ab}{x+b}$ Replace y with $y-a$ $\longrightarrow$ $y=\dfrac{4+ab}{x+b}$ Replace x with $x-b$ $\longrightarrow$ $y=\dfrac{4+ab}{x}$ Replace y with $y(4+ab)$ $\longrightarrow$ $y=\dfrac{1}{x}$ OR:Replace x with $x(4+ab)$ 1. Translate by a units in the positive y direction. 2. Translate by b units in the positive x direction. 3. Scale parallel to y-axis (or x-axis) with a factor of $\dfrac{1}{4+ab}$. **OR** $y=-a+\dfrac{4+ab}{x+b}$ Replace y with $y-a$ $\longrightarrow$ $y=\dfrac{4+ab}{x+b}$ Replace y with $y(4+ab)$ $\longrightarrow$ $y=\dfrac{1}{x+b}$ Replace x with $x-b$ $\longrightarrow$ $y=\dfrac{1}{x}$ 1. Translate by a units in the positive y direction. 2. Scale parallel to y-axis with a factor of $\dfrac{1}{4+ab}$. 3. Translate by b units in the positive x direction.	Use the idea of replacement to help you decide the sequence of transformations. Use the correct terminology to describe the transformations.

(c)	$y=\dfrac{4-ax}{x+b}$ $xy+by=4-ax$ $x=\dfrac{4-by}{y+a}$ $\therefore \mathrm{f}^{-1}(x)=\dfrac{4-bx}{x+a}$ $\mathrm{R}_{\mathrm{f}^{-1}}=\mathrm{D}_{\mathrm{f}}=\mathbb{R}\setminus\{-b\}$ **OR** $\{x\in\mathbb{R}\mid x\neq -b\}$ **OR** $(-\infty,-b)\cup(-b,\infty)$	
(d)	If $b=a$, $\mathrm{f}(x)=\dfrac{4-ax}{x+a}=\mathrm{f}^{-1}(x)$ Then $\mathrm{f}^2(x)=\mathrm{ff}(x)=\mathrm{ff}^{-1}(x)=x$ $\mathrm{f}^4(x)=\mathrm{f}^2\left[\mathrm{f}^2(x)\right]=\mathrm{f}^2(x)=x$ Continuing, $\mathrm{f}^{2024}(x)=x$ $\therefore \mathrm{f}^{2025}(2)=\mathrm{f}\left[\mathrm{f}^{2024}(2)\right]=\mathrm{f}(2)=\dfrac{4-2a}{2+a}$	

Question 11

No.	Suggested Solution	Remarks for Student
(a)	$x=28(\theta\cos\theta+1)\Rightarrow\dfrac{\mathrm{d}x}{\mathrm{d}\theta}=28(\cos\theta-\theta\sin\theta)$ $y=20(2-\theta\sin\theta)\Rightarrow\dfrac{\mathrm{d}y}{\mathrm{d}\theta}=20(-\sin\theta-\theta\cos\theta)$ $\therefore\dfrac{\mathrm{d}y}{\mathrm{d}x}=\dfrac{\frac{\mathrm{d}y}{\mathrm{d}\theta}}{\frac{\mathrm{d}x}{\mathrm{d}\theta}}=\dfrac{20(-\sin\theta-\theta\cos\theta)}{28(\cos\theta-\theta\sin\theta)}=\dfrac{5(\theta\cos\theta+\sin\theta)}{7(\theta\sin\theta-\cos\theta)}$	
(b)(i)	At C and D, $\dfrac{\mathrm{d}y}{\mathrm{d}x}=\dfrac{5(\theta\cos\theta+\sin\theta)}{7(\theta\sin\theta-\cos\theta)}$ is undefined. Therefore, $\theta\sin\theta-\cos\theta=0$ Using GC, $\theta=-0.860334$ or 0.860334 $\theta=-0.86$ or 0.86 (2 s.f.)	GC Screen shots:  

(b)(ii)	When $\theta = 0.860334$, $x \approx 12.289$ When $\theta = -0.860334$, $x \approx 43.711$ The greatest width, CD, of the loop $= 43.711 - 12.289 = 31.4$ metres (3 s.f.)	
(c)(i)	x-coordinate of point A is 28 $\Rightarrow 28(\theta\cos\theta + 1) = 28$ $\Rightarrow \theta\cos\theta + 1 = 1$ $\Rightarrow \theta\cos\theta = 0$ $\Rightarrow \theta = 0$ or $\cos\theta = 0$ $\Rightarrow \theta = 0$ or $\theta = -\frac{\pi}{2}$ or $\frac{\pi}{2}$ When $\theta = 0$, $y = 20(2-0) = 40$ When $\theta = -\frac{\pi}{2}$, $y = 20\left(2 + \frac{\pi}{2}(-1)\right) = 40 - 10\pi < 40$ When $\theta = \frac{\pi}{2}$, $y = 20\left(2 - \frac{\pi}{2}(1)\right) = 40 - 10\pi < 40$ Clearly, at point A, $\theta = 0$.	
(c)(ii)	The 2 possible values of θ at point B are $-\frac{\pi}{2}$ and $\frac{\pi}{2}$.	
(c)(iii)	Distance $AB = 40 - (40 - 10\pi) = 10\pi = 31.4$ metres (3 s.f.)	
(d)	$CD \approx 31.422$ $AE = 40 - 20(2 - (0.860334)\sin(0.860334)) \approx 13.044$ 90% of $\frac{1}{2}CD = 90\% \times \frac{31.422}{2} \approx 14.140$ 110% of $\frac{1}{2}CD = 110\% \times \frac{31.422}{2} \approx 17.282$ Since AE does not lie within $(14.140, 17.282)$, the design of this section of track does not satisfy the condition.	

Question 12

No.	Suggested Solution	Remarks for Student
(a)	Since all the coefficients of the polynomial are real, **non-real** roots occur in conjugate pairs. Given that $1+2\text{i}$ is a root of this equation, then $1-2\text{i}$ is a also root.	
(b)	Method 1: $\left[z-(1+2\text{i})\right]\left[z-(1-2\text{i})\right]$ $=\left[(z-1)-2\text{i}\right]\left[(z-1)+2\text{i}\right]$ $=z^2-2z+5$ $\therefore z^3+pz^2+qz-10=(z^2-2z+5)(z-k)$ where k is real Compare coefficients of z^0 : $k=2$ z^2 : $p=-2-2=-4$ z : $q=5+2(2)=9$ $\therefore$ The third root is $z=2$. Method 2: $(1+2\text{i})^2=-3+4\text{i}$ $(1+2\text{i})^3=(1+2\text{i})^2(1+2\text{i})=(-3+4\text{i})(1+2\text{i})=-11-2\text{i}$ Now $(1+2\text{i})^3+p(1+2\text{i})^2+q(1+2\text{i})-10=0$ $(-11-2\text{i})+p(-3+4\text{i})+q(1+2\text{i})-10=0$ Compare real and imaginary parts: Real part : $-11-3p+q-10=0\Rightarrow -3p+q=21$ --- (1) Imaginary part : $-2+4p+2q=0\Rightarrow 4p+2q=2$ --- (2) $(2)-2\times(1)\Rightarrow 10p=-40\Rightarrow p=-4\ \Rightarrow q=21+3(-4)=9$ $\left[z-(1+2\text{i})\right]\left[z-(1-2\text{i})\right]$ $=\left[(z-1)-2\text{i}\right]\left[(z-1)+2\text{i}\right]$ $=z^2-2z+5$ $\therefore z^3-4z^2+9z-10=(z^2-2z+5)(z-k)$ where k is real By observation, $k=2$, so the third root is $z=2$	Method 1 is clearly more efficient. If you choose to use method 2, do note that the calculator is **not** allowed. Working for the various parts must be shown clearly. Do use the GC to check your answer nevertheless.

(c)	Note that $w = 0$ is not a solution to the equation. $1 + pw + qw^2 - 10w^3 = 0$ Dividing throughout by w^3 $\left(\frac{1}{w}\right)^3 + p\left(\frac{1}{w}\right)^2 + q\left(\frac{1}{w}\right) - 10 = 0$ Replace z with $\frac{1}{w}$, $\frac{1}{w} = 1 + 2\text{i}$ or $\frac{1}{w} = 1 - 2\text{i}$ or $\frac{1}{w} = 2$ $w = \frac{1}{1+2\text{i}} \times \frac{1-2\text{i}}{1-2\text{i}}$ $w = \frac{1}{1-2\text{i}} \times \frac{1+2\text{i}}{1+2\text{i}}$ $w = \frac{1}{2}$ $= \frac{1}{5}(1-2\text{i})$ $= \frac{1}{5}(1+2\text{i})$	Note that calculator is **not** allowed. Hence, show all working clearly!